

12

Equilibria, rates, and mechanisms

Connections

➡ Building on

- Structure of molecules [ch4](#)
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- Nucleophilic attack on carbonyl groups [ch6 & ch9](#)
- Acidity and pK_a [ch8](#)
- Substitutions at carbonyl groups [ch11 & ch12](#)

Arriving at

- What controls equilibria
- Free energy, enthalpy, and entropy
- What controls the rates of reactions
- Intermediates and transition states
- How catalysts work
- Effects of temperature on reactions
- Why the solvent matters
- Rate equations and their link to mechanism

➡ Looking forward to

- Substitution reactions at saturated C [ch15](#)
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- How mechanisms are discovered [ch39](#)

'One could no longer just mix things; sophistication in physical chemistry was the base from which all chemists—including the organic—must start.'
Christopher Ingold (1893–1970). Ingold uncovered many of the mechanisms we now take for granted in organic chemistry.

If you go into a chemistry laboratory, you will see some reactions being heated in boiling solvent (perhaps 80 to 120 °C), and you will see others being performed at maybe –80 °C or below. Some reactions are over in a few minutes; others are left for hours. In some reactions the amounts of reagents are critical; in others large excesses are used. Some reactions use water as a solvent; in others it must be rigorously excluded, and perhaps toluene, ether, ethanol, or DMF is essential for the success of the reaction. Why such a diverse range of conditions? How can conditions be chosen to favour the reaction we want? To explain all this we will need to work through some thermodynamic principles. We will take a practical, visual approach to the topic, and we will avoid detailed algebraic discussion: for that you are welcome to turn to a textbook of physical chemistry—there are some suggestions at the end of this chapter. In fact, we will use only two algebraic equations. Both are so important that you should memorize them; the second in particular can be extremely valuable when we think about how to get reactions to work.

How far and how fast?

In previous chapters we have said things about the *reversibility of reactions*:

'Cyanohydrin formation is reversible: just dissolving a cyanohydrin in water can give back the aldehyde or ketone you started with' (Chapter 6); 'HCl transfers its proton almost completely to water, and is a strong acid. But the transfer of protons to water from carboxylic acids is only partial' (Chapter 8); 'This step is irreversible because SO₂ and HCl are gases that are lost from the reaction mixture' (Chapter 10); 'The tetrahedral intermediate can collapse either way, giving back ester or going forward to acid plus alcohol.' (Chapter 10);

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about the *relative stability* of different compounds:

‘The most important factor in the strength of an acid is the stability of the conjugate base’ (Chapter 8); ‘F⁻ is much more stable than CH₃⁻ because fluorine is much more electronegative than carbon’ (Chapter 8); ‘Oximes are more stable than imines because the electronegative substituent can participate in delocalization of the imine double bond’ (Chapter 11);

and about the *rate of reactions*:

‘Benzaldehyde is reduced about 400 times faster than acetophenone in isopropanol’ (Chapter 6); ‘While amines react with acetic anhydride quite rapidly at room temperature (reaction complete in a few hours), alcohols react extremely slowly in the absence of a base’ (Chapter 10); ‘Secondary and tertiary amides are difficult to hydrolyse but a similar mechanism is successful with only a little water and plenty of a strong base’ (Chapter 10); ‘Acyclic hemiacetals form relatively slowly from an aldehyde or ketone plus an alcohol, but their rate of formation is greatly increased either by acid or by base’ (Chapter 11).

We are now going to consider in detail why some reactions can run forwards or backwards, why some form products irreversibly, why some reach an equilibrium, why some reactions go fast and some go slow, and what stability has to do with all of this. Understanding these factors will allow you to make the reactions you want to happen go faster and the reactions you don’t want to happen go slower, giving you a product in a useful yield. We shall be breaking reaction mechanisms down into steps and working out which step is the most important. But first we must consider what we really mean by the ‘stability’ of molecules and what determines how much of one substance you get when it is in equilibrium with another.

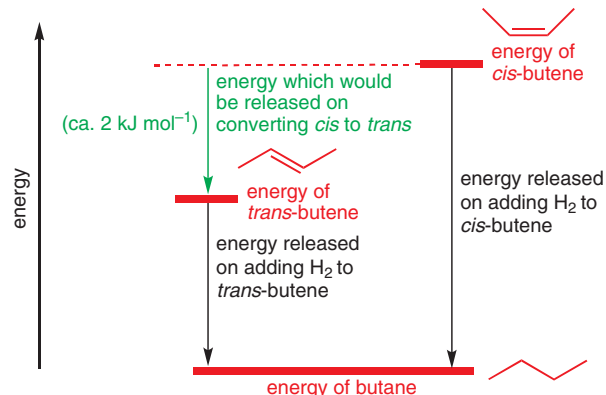
➡ We looked at the problem of how to make ketones from esters by increasing the rate of one reaction at the expense of another in Chapter 10, p. 218.

Stability and energy levels

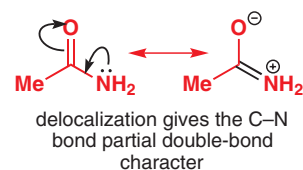
So far we have been rather vague about the term ‘stability’, just saying things like ‘this compound is more stable than that compound’. What we really mean is that this compound has **less energy** than that one. For example, as you know from Chapters 4 and 7 alkenes can come in two forms we can call *cis* and *trans*. In general *trans*-alkenes are more stable than *cis*-alkenes. How do we know? Well, we can convert both *cis*- and *trans*-butene to the same alkane, butane, by adding a molecule of hydrogen. Energy is given out during the reaction, and if we measure how much energy we get from hydrogenation of *trans*-butene and compare it with the amount we get from *cis*-butene, we find that the *cis*-alkene give us about 2 kJ mol⁻¹ more. *Cis*-butene is higher in energy, and must therefore be less stable. We can represent this in the energy profile diagram on the right. The two red lines show the energies of the molecules, and the black arrows the amount of energy released when hydrogen is added.

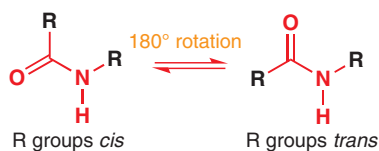
This comparison of energy is most interesting when two compounds can interconvert. For example, as you saw in Chapter 7, rotation about the C–N bond of an amide is slow because delocalization of the N lone pair gives it some double-bond character.

The C–N bond can rotate, but the rotation is slow and can be measured by NMR spectroscopy. We might expect to find two forms of an amide of the type RNH–COR: one with the two R groups *trans* to one another, and one with them *cis*. Depending on the size of R we should expect one form to be more stable than the other and we can represent this on an energy profile diagram showing the relationship between the two molecules in energy terms.

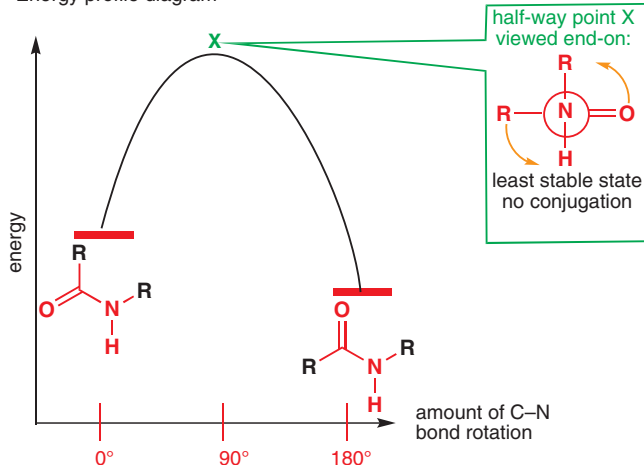


➡ We used a similar argument to compare the stabilities of benzene and cyclooctatetraene (see p. 157).





Energy profile diagram



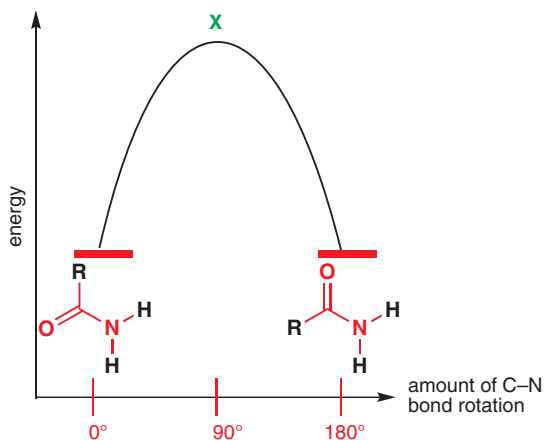
■ *Cis*- and *trans*-alkenes don't usually interconvert without catalysis. You can read more about this on p. 105.

This time there is an axis along the bottom indicating the extent of rotation about the C–N bond. The two red lines show the energies of the molecules and the curved black line shows what must happen in energy terms as the two forms interconvert. Energy goes up as the C–N bond starts to rotate and reaches a maximum at point X when rotation by 90° has removed the conjugation (the nitrogen lone pair can't delocalize into the C=O bond because it is perpendicular to the C=O π^* orbital) before falling again as the conjugation is regained.

The relative energies of the two states will depend on the nature of R. The situation we have shown, with the *cis* arrangement being much less stable than the *trans*, would apply to large R groups. We can define an equilibrium constant K for this process. For large R groups, K will be very large:

$$K = \frac{[\text{amide with R groups } \textit{trans}]}{[\text{amide with R groups } \textit{cis}]}$$

At the other extreme is the case when both substituents on nitrogen are H. Then the two arrangements would have equal energies. The process which interconverts the structures is the same but there is now no difference between them. If you could measure an equilibrium constant, it would now be exactly $K = 1$.



In more general terms, amide rotation is a simple example of an equilibrium reaction. If we replace 'amount of C–N bond rotation' with 'reaction coordinate' we have a picture of a typical reaction in which reagents and products are in equilibrium.

How the equilibrium constant varies with the difference in energy between reactants and products

You saw that when the energies of the two forms of the amide were the same, the equilibrium constant for their interconversion must be $K = 1$. When one was higher in energy than the other, we just said that K was 'large'. But we can be more specific. For any reaction in equilibrium, the equilibrium constant K is related to the difference in energy between the starting materials and the products by the following equation:

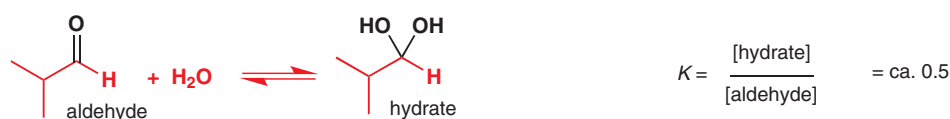
$$\Delta G = -RT \ln K$$

where ΔG (the **free energy** of the reaction) is the difference in energy between the two states (in kJ mol^{-1}), T is the temperature (in kelvin, not $^{\circ}\text{C}$), and R is a constant known as the **gas constant** and equal to $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$.

This equation tells us that we can work out the **equilibrium composition** (how much of each component there is at equilibrium) provided we know the difference in energy between the products and reactants.

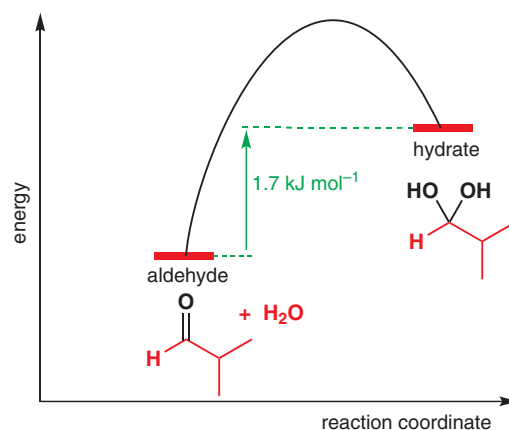
An example: hydration of an aldehyde

In Chapter 6 we showed you that water adds reversibly to the carbonyl group of an aldehyde: the aldehyde and the hydrate are in equilibrium. Here's the example with isobutyraldehyde (2-methylpropanal). The equilibrium constant is the concentration of hydrate at equilibrium divided by the concentration of aldehyde, also at equilibrium.



The concentrations of hydrate and aldehyde at equilibrium in water may be determined by measuring the UV absorption of known concentrations of aldehyde in water and comparing these with the absorptions in a solvent such as cyclohexane where no hydrate formation is possible. Such experiments reveal that the equilibrium constant for this reaction in water at 25°C is approximately 0.5 so that there is about twice as much aldehyde as hydrate in the equilibrium mixture.

Using the equation above, we find that the corresponding value for ΔG is $-8.314 \times 298 \times \ln(0.5) = +1.7 \text{ kJ mol}^{-1}$. In other words, the solution of the hydrate in water is 1.7 kJ mol^{-1} higher in energy than the solution of the aldehyde in water. All this can be shown on an energy profile diagram.



The sign of ΔG tells us whether products or reactants are favoured at equilibrium

In the equilibrium above, the hydrate is higher in energy than the aldehyde: at equilibrium there is more aldehyde than hydrate, and the equilibrium constant is therefore less than 1. Whenever this is the case (i.e. the equilibrium lies to the side of the reactants, rather than the

■ The *reaction coordinate* is simply an arbitrary measure of the progress of a molecule of starting material as it turns into a molecule of product. You will see it in several diagrams in this chapter.

This relationship was derived by the American physical chemist J. Willard Gibbs in the 1870s.

■ Although water is involved in the reaction, you saw on p. 169 that the concentration of neat water effectively remains constant at 55.5 mol dm^{-3} and is usually not included in the equilibrium constant.

■ The *reaction coordinate* is an arbitrary scale used for diagrammatic purposes only.

■ The sign of ΔG for a reaction tells us whether the starting materials or products are favoured at equilibrium, but it tells us nothing about how long it will take before equilibrium is reached. The reaction could take hundreds of years! This will be dealt with later.

products) K will be less than 1. This means that its logarithm must be negative and, because $\Delta G = -RT \ln K$, ΔG must be positive. Conversely, for a reaction in which products are favoured over reactants, K must be greater than 1, its logarithm will be positive, and hence ΔG must be negative. When K is exactly 1, since $\ln 1 = 0$, ΔG will be zero.

● **ΔG tells us about the position of equilibrium.**

- If ΔG for a reaction is negative, the products will be favoured at equilibrium.
- If ΔG for a reaction is positive, the reactants will be favoured at equilibrium.
- If ΔG for a reaction is zero, the equilibrium constant for the reaction will be 1.

A small change in ΔG makes a big difference in K

The tiny difference in energy between the hydrate and the aldehyde (1.7 kJ mol^{-1} is small: the strength of a typical C—C bond is about 350 kJ mol^{-1}) gave an appreciable difference in the equilibrium composition. This is because of the logarithm term in the equation $\Delta G = -RT \ln K$: relatively small energy differences have a very large effect on K . The table below shows the equilibrium constants, K , that correspond to energy differences, ΔG , between 0 and 50 kJ mol^{-1} . These are relatively small energy differences, but the equilibrium constants change by enormous amounts.

Energies in older or American books are sometimes quoted in kcal (kilocalories) mol^{-1} . $1 \text{ kcal} = 4.184 \text{ kJ}$. The 'calories' counted by nutritionists are in fact kilocalories; the typical energy output of a human adult is 10,000 kJ per day.

Variation of K with ΔG

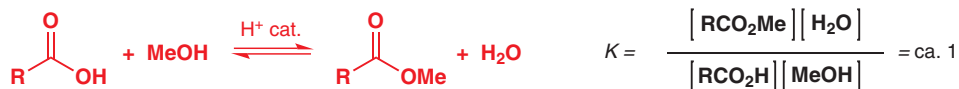
$\Delta G, \text{ kJ mol}^{-1}$	K	% of more stable state at equilibrium
0	1.0	50
1	1.5	60
2	2.2	69
3	3.5	77
4	5.0	83
5	7.5	88
10	57	98
15	430	99.8
20	3200	99.97
50	580 000 000	99.999998

In a typical chemical reaction, 'driving an equilibrium over to products' might mean getting, say, 98% of the products and only 2% of starting materials. You can see in the table that this requires an equilibrium constant of just over 50 and an energy difference of only 10 kJ mol^{-1} . This small energy difference is quite enough—after all, a yield of 98% is rather good!

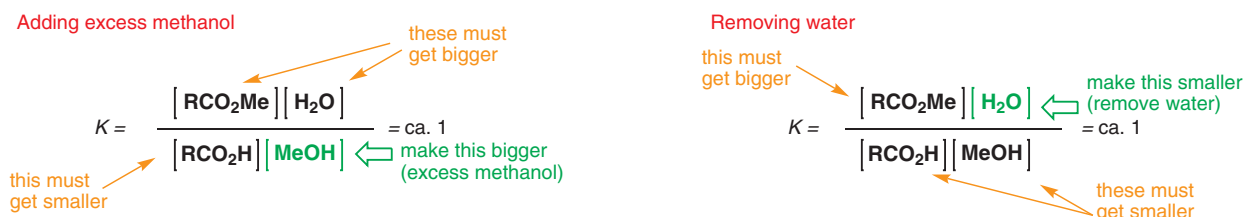
How to make the equilibrium favour the product you want

The direct formation of esters

The formation and hydrolysis of esters was discussed in Chapter 10 where we established that acid and ester are in equilibrium and that the equilibrium constant is about 1. Since the position of the equilibrium favours neither the starting materials nor the products, how can we manipulate the conditions of the reaction if we actually want to make 100% ester?



The important point is that, at any one particular temperature, the equilibrium constant is just that—*constant*. This gives us a means of forcing the equilibrium to favour the products (or reactants) since the ratio between them must remain constant. Imagine what happens if we add more methanol to the reaction above. $[\text{MeOH}]$ increases, but the overall value of K has to stay the same. The only way this can happen is if more of the ester converts to the acid. Alternatively, imagine removing water from the equilibrium. $[\text{H}_2\text{O}]$ goes down, so to bring K back to the value of 1, the concentrations of acid and methanol are going to have to go down too, by converting themselves to ester and water.

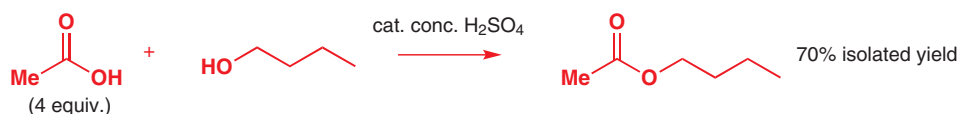


This is exactly how the equilibrium is manipulated in practice. One way to make esters in the laboratory is to use a large excess of the alcohol and remove water continually from the system as it is formed, for example by distilling it out. This means that in the equilibrium mixture there is a tiny quantity of water, lots of the ester, lots of the alcohol, and very little of the carboxylic acid; in other words, we have converted the carboxylic acid into the ester. We must still use an acid catalyst, but the acid must be anhydrous since we do not want any water present—commonly used acids are toluenesulfonic acid (tosic acid, TsOH), concentrated sulfuric acid (H_2SO_4), or gaseous HCl . The acid catalyst does not alter the position of the equilibrium; it simply speeds up the rate of the reaction, allowing equilibrium to be reached more quickly. This is an important point that we will come back to shortly.

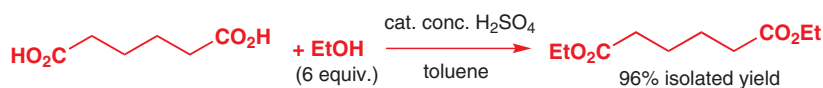
➡ There is more on TsOH on p. 227.

Typical method for making an ester

Reflux the carboxylic acid with an excess of the alcohol (or the alcohol with an excess of the carboxylic acid) with about 3–5% of a mineral acid (usually HCl or H_2SO_4) as a catalyst and distil out the water that is formed in the reaction. For example, butanol was heated under reflux with a fourfold excess of acetic acid and a catalytic amount of concentrated H_2SO_4 to give butyl acetate in a yield of 70%.



It may also help to distil out the water that is formed in the reaction: diethyl adipate (the diethyl ester of hexanedioic acid) can be made in toluene solution using a sixfold excess of ethanol, concentrated H_2SO_4 as catalyst, heating in toluene, and distilling out the water using a Dean Stark apparatus. You can tell from the yield that the equilibrium is very favourable.



In these cases the equilibrium is made more favourable by using an excess of reagents and/or removing one of the products. The equilibrium *constant* remains the same.

■ 'Reflux' means boil underneath a condenser, so that the boiling solvent constantly runs back into the reaction and is not lost.

➡ The Dean Stark apparatus for removing water from a refluxing mixture is described on p. 228.

■ The high temperatures and acid catalysis are used to speed up *arrival* at equilibrium, which would otherwise take several days. This aspect of reactivity—the rate of the reaction, rather than the position of the equilibrium—will be dealt with shortly.

Typical method for hydrolysing an ester

Almost all methods for hydrolysing an ester in order to convert it back to an acid and an alcohol simply make use of excess water. Increasing $[\text{H}_2\text{O}]$ forces more acid and alcohol to form to restore the equilibrium, and in favourable situations high yields of the acid and alcohol are formed.

Entropy is important in determining equilibrium constants

The equation we introduced on p. 243 tells us that an equilibrium favours whichever of the reactants or products has lower energy. But you might reasonably ask this question: why does it just *favour* the components with lower energy? Why do you get *any* of the higher energy ones at all? For the hydration on p. 243, for example, the hydrate is 1.7 kJ mol^{-1} higher in energy than the starting aldehyde, so why does the aldehyde react at all? Surely the equilibrium would attain a lower energy state, not with just an excess of aldehyde over hydrate, but with no hydrate at all?

■ That equation again:
 $\Delta G = -RT \ln K$

The answer is due to *entropy*, a measure of disorder. Even when there is a difference in energy between the starting materials and products in an equilibrium, you still get *some* of the less stable components. Put simply, having a mixture of components is favourable because a mixture has higher entropy than a pure compound, and equilibria tend to maximize overall entropy. This may be quite a new concept to you, so we will now work our way stepwise through these ideas.

Energy, enthalpy, and entropy: ΔG , ΔH , and ΔS

The equation in the margin just above tells us that the sign and magnitude of the energy ΔG are the only things that matter in deciding whether an equilibrium goes in one direction or another. If ΔG is negative the equilibrium will favour the products and if ΔG is large and negative the reaction can go to completion. The table on p. 244 tells us that it is enough for ΔG to be only about -10 kJ mol^{-1} to get complete reaction. But we haven't yet considered what ΔG actually corresponds to *physically*.

To do this we need to introduce our second equation. The free energy of a reaction, ΔG , is related to two other quantities, the enthalpy of reaction, ΔH , and the entropy of reaction, ΔS , by the equation:

$$\Delta G = \Delta H - T\Delta S$$

■ If you are interested in the derivation of this equation, which is an expression of the second law of thermodynamics, you will need to consult a textbook of physical chemistry. But you will be able to follow the explanations below without knowing the background to the equation.

As before, T is the temperature of the reaction in kelvin. Enthalpy, H , is a measure of heat, and the change in enthalpy, ΔH , in a chemical reaction is the **heat given out or taken up** in that reaction. Reactions which give out heat are called *exothermic*, and have negative ΔH ; reactions which take in heat are called *endothermic* and have positive ΔH . Since breaking bonds requires energy and making bonds liberates energy, the enthalpy change gives an indication of whether the products have more stable bonds than the starting materials or not.

Entropy, S , is a measure of the **disorder in the system**, so ΔS represents the entropy difference—the change in disorder—between the starting materials and the products. More disorder gives a positive ΔS ; less disorder a negative ΔS .

So ΔG represents a combination of heat and disorder. But what does this mean for you as a chemist wanting to get a reaction to work the way you want it to? We know that for a favourable change (i.e. an equilibrium favouring products) ΔG must be negative—in fact the more negative the better, as this gives a larger equilibrium constant. Since $\Delta G = \Delta H - T\Delta S$, we get a large, negative ΔG most readily if:

1. ΔH is negative, i.e. the reaction is exothermic.

and

2. ΔS is positive (and hence $-T\Delta S$ is negative), i.e. the reaction becomes more disordered.

Of course, we can still get a negative ΔG from an endothermic reaction (i.e. from a positive ΔH) but only if the reaction products are more disordered than the starting materials; likewise a reaction which becomes more ordered as it proceeds can still be favourable, but only if it is exothermic to compensate for the loss of entropy.

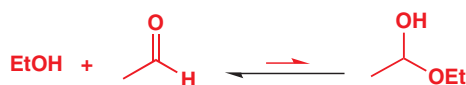
Because of the factor T multiplying the entropy term, both the equilibrium constant K (which depends on ΔG) and the relative importance of the two quantities (ΔH and ΔS) will vary with temperature (entropy changes are more important at higher temperatures). We'll now look at some examples to see how this works in practice.

Enthalpy versus entropy—some examples

Entropy dominates equilibrium constants in the difference between inter- and intramolecular reactions. In Chapter 6 we explained that hemiacetal formation is often an equilibrium, with neither starting materials nor products strongly favoured. The addition of ethanol to acetaldehyde shown below on the left, for example, has an equilibrium constant not far from 1. Overall, ΔG must therefore be approximately 0 (in fact it's very slightly positive). The *enthalpy* change associated with the reaction is the result of the change in bonding: in this case, a C=O double bond becomes two C–O single bonds, and these two single bonds are marginally more stable than the C=O double bond, therefore ΔH is slightly negative. But working against this is the fact that every molecule of hemiacetal that forms consumes two molecules of starting material. Decreasing the number of molecules (and moving from a mixture of aldehyde and alcohol towards pure hemiacetal) leads to an increase in the order of the mixture—in other words a *decrease* in entropy. ΔS is negative, so the $-T\Delta S$ is positive, just about counterbalancing the small negative ΔH , and giving a slightly positive ΔG .

A mixture has more entropy than a pure substance because there are many more ways of arranging a mixture. Imagine lining up every molecule in a mole of substance and a mole of a 1:1 mixture. For the pure substance, each member of the line of molecules has to be the same. For the mixture, at every position in the line there is a choice of two alternatives, giving a huge number of possible arrangements.

intermolecular hemiacetal formation

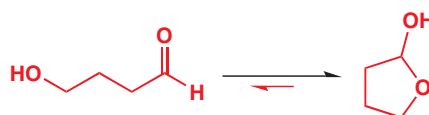


ΔH is small and negative because C=O double bond is slightly less stable than 2 x C–O single bonds

ΔS is negative because the one molecule of product is intrinsically less disordered than the two molecules of starting material

Since $\Delta G = \Delta H - T\Delta S$, ΔG is positive and the equilibrium lies to the left

intramolecular hemiacetal formation



ΔH is again small and negative because C=O double bond is slightly less stable than 2 x C–O single bonds

ΔS is no longer negative; there is no decrease in the number of molecules in this reaction

Since $\Delta G = \Delta H - T\Delta S$, ΔG is negative and the equilibrium lies to the right

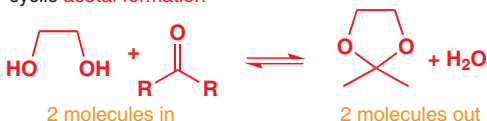
The reaction on the right is different because it is an *intramolecular* reaction: the hydroxyl group and aldehyde lie in the same molecule. ΔH will have essentially the same value as in the intermolecular reaction on the left, but as the intramolecular reaction progresses, one molecule stays one molecule—there is consequently a much less significant decrease in entropy. Our $T\Delta S$ term no longer weighs against the negative ΔH term, making ΔG negative overall and allowing the equilibrium to lie to the right.

In Chapter 11 we showed you how acetals can be used as base-stable protective groups to prevent nucleophiles attacking carbonyl groups. The acetals we chose to use were cyclic compounds known as dioxolanes, for a very good reason: cyclic acetals are more resistant to hydrolysis than their acyclic counterparts. They are also easier to make—they form quite readily, even from ketones. Again, we have entropic factors to thank for their stability. For the formation of an acyclic acetal (below on the left), three molecules go in and two come out, but for a cyclic one, a cyclic acetal, two molecules go in (ketone plus diol) and two molecules come out (acetal plus water), so the usually unfavourable ΔS factor is no longer against us.

acyclic acetal formation



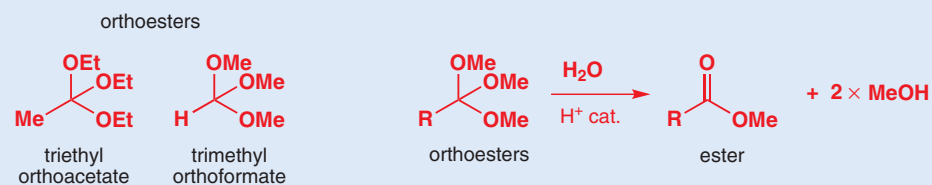
cyclic acetal formation



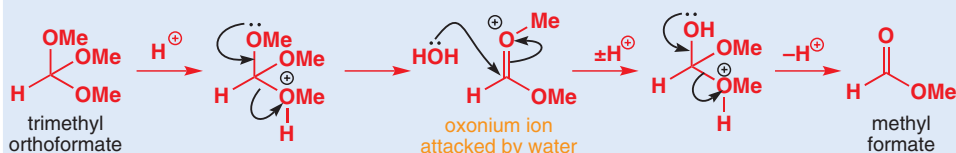
➡ Look back at p. 227 to remind yourself of this.

Overcoming entropy: orthoesters

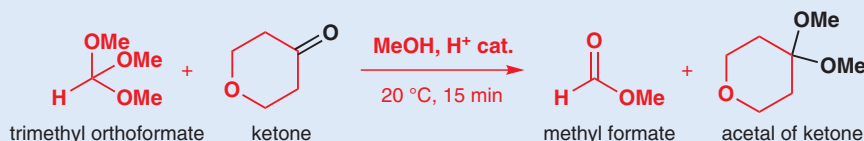
There is a neat way of sidestepping the entropic problem associated with making acyclic acetals: we can use an *ortho*-ester as a source of alcohol. Orthoesters can be viewed as the 'acetals of esters', which are hydrolysed by water, when catalysed by acid, to an ordinary ester and two molecules of alcohol.



Here is the mechanism for the hydrolysis—you should be feeling quite familiar with this sort of thing by now.



Ketones or aldehydes undergo acetal exchange with orthoesters. The mechanism starts off as if the orthoester is going to hydrolyse but the alcohol released adds to the ketone and acetal formation begins. The water produced is taken out of the equilibrium by hydrolysis of the orthoester, and we get two molecules from two: entropy is no longer our enemy.



Equilibrium constants vary with temperature

We have said (p. 245) that the equilibrium constant is a constant only as long as the temperature does not change. We can work out exactly how the equilibrium constant varies with temperature by putting our two all-important equations $\Delta G = -RT \ln K$ and $\Delta G = \Delta H - T\Delta S$ together to make

$$-RT \ln K = \Delta H - T\Delta S$$

If we divide throughout by $-RT$ we have

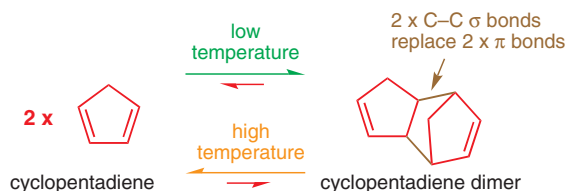
$$\ln K = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

This equation separates the equilibrium constant K into enthalpy and entropy terms, but it is the enthalpy term that determines how K varies with temperature. Plotting $\ln K$ against $1/T$ would give us a straight line with slope $-\Delta H/R$ and intercept $\Delta S/R$. Since T (the temperature in kelvin) is always positive, whether the slope is positive or negative depends on the sign of ΔH : if ΔH is negative then, as temperature increases, $\ln K$ (and hence K) increases. In other words, if the reaction is exothermic (that is, gives out heat) then at higher temperatures the equilibrium constant will be smaller. For an endothermic reaction, as the temperature is increased, the equilibrium constant increases.

Some reactions are reversible on heating: cracking

Notice that the equation above also tells us that enthalpy becomes a less important contributor to the equilibrium constant as temperature increases, so the higher the temperature, the more important is the entropy term. This fact means that some reactions favour one side of the equilibrium at low temperature but the other at high temperature. Here is an example: the dimerization of cyclopentadiene. You will meet the mechanism of this reaction in Chapter 34, but for

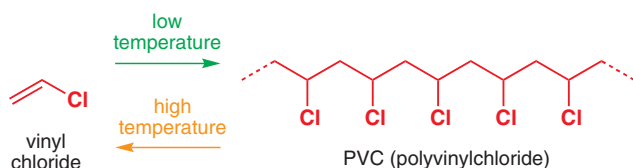
now we can just treat it as a simple dimerization reaction in which two C=C π bonds are replaced by two C—C σ bonds—enthalpically a very favourable process because σ bonds are stronger than π bonds. On standing at low temperature, cyclopentadiene converts to the dimer even though two monomer molecules have more entropy than one molecule of the dimer.



But on heating, the dimer breaks down to give monomeric cyclopentadiene: the equilibrium constant now favours the starting materials. As we predicted, because the reaction is exothermic, heating it makes it less favourable. You can also think of it in terms of our earlier equation $\Delta G = \Delta H - T\Delta S$: at low temperature, the large negative ΔH term dominates, and ΔG is large and negative too. But as T increases, the positive ΔS becomes more important, and eventually $T\Delta S$ overtakes ΔH and ΔG becomes positive, and the reaction now favours starting materials.

If you want to use cyclopentadiene, you have to heat the dimer to ‘crack’ it (‘cracking’ is the term used for getting monomers from dimers or polymers). If you lazily leave the monomer overnight and plan to do your reaction tomorrow, you will return in the morning to find dimer.

This idea becomes even more pointed when we look at polymerization. Polyvinyl chloride is the familiar plastic PVC and is made by reaction of large numbers of monomeric vinyl chloride molecules. There is, of course, an enormous decrease in entropy in this reaction any polymerization will not occur above a certain temperature. Some polymers can be depolymerized at high temperatures and this can be the basis for recycling.



Everything decomposes at a high enough temperature eventually, giving atoms. This is because the entropy for lots of particles all mixed up is much greater than that of fewer larger particles.

● Summary: Practical points from thermodynamic theory

- The free energy change ΔG in a reaction is proportional to $\ln K$ (that is, $\Delta G = -RT \ln K$).
- ΔG and K are made up of enthalpy and entropy terms (that is, $\Delta G = \Delta H - T\Delta S$).
- The enthalpy change ΔH is the difference in stability (bond strength) of the reagents and products.
- The entropy change ΔS is the difference between the disorder of the reagents and that of the products.
- The enthalpy term alone determines how K varies with temperature.
- The entropy change come to dominate control of equilibrium as temperature is raised.

Le Châtelier's principle

You may well be familiar with a rule that helps to predict how a system at equilibrium responds to a change in external conditions—**Le Châtelier's principle**. This says that if we disturb a system at equilibrium it will respond so as to minimize the effect of the disturbance. An example of a disturbance is adding more starting material to a reaction mixture at equilibrium. What happens? More product is formed to use up this extra material. This is a consequence of the equilibrium constant being, well... constant and hardly needs anybody's principle.

Another disturbance is heating. If a reaction under equilibrium is heated, how the equilibrium changes depends on whether the reaction is exothermic or endothermic. If it is exothermic (that is, gives out heat), Le Châtelier's principle would predict that, since heat is consumed in the *reverse* reaction, more of the starting materials will be formed. Again no 'principle' is needed—this change occurs because the equilibrium constant is smaller at higher temperatures in an exothermic reaction. Avoid using principles and rules without understanding the science underneath them or you may find yourself playing with fire (which incidentally most definitely does not obey Le Châtelier's principle, for very good reasons...).

Introducing kinetics: how to make reactions go faster and cleaner

Although in chemistry laboratories you will see lots of reactions being heated, very rarely will this be to alter the equilibrium position. This is because most reactions are not carried out reversibly and so the ratio of products to reactants is not an equilibrium ratio. The main reason chemists heat up reactions is simple—it speeds them up. The study of the rates of reactions, as opposed to their equilibrium states, is known as **kinetics**.

● Thermodynamics is concerned with equilibria; kinetics is concerned with rates.

How fast do reactions go? Activation energies

The combustion of the hydrocarbon shown below, the major component of petrol (gasoline) trivially known as ‘isooctane’, proceeds with $\Delta G = -1000 \text{ kJ mol}^{-1}$ at 298 K.

■ ‘Isooctane’ is the trivial name of 2,2,4-trimethylpentane.



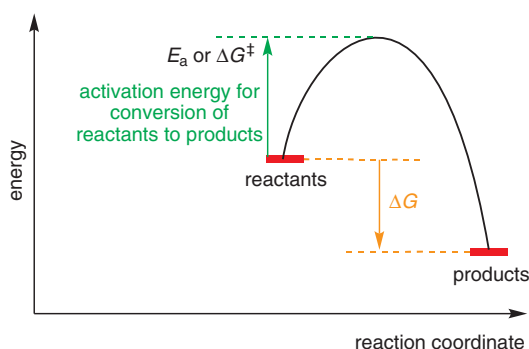
The table on p. 244 shows that even a ΔG of only -50 kJ mol^{-1} gives rise to a huge equilibrium constant: $-1000 \text{ kJ mol}^{-1}$ gives an equilibrium constant of 10^{175} (at 298 K), a number too vast to contemplate (there are ‘only’ about 10^{86} atoms in the observable universe). This value of ΔG (or the corresponding value for the equilibrium constant) suggests that isooctane simply could not exist in the presence of oxygen. Yet we put it into the fuel tanks of our cars every day—clearly something is wrong.

Since isooctane can exist in an atmosphere of oxygen despite the fact that the equilibrium position really would be completely on the side of the combustion products, the only conclusion we can draw must be that a mixture of isooctane and oxygen cannot be at equilibrium. A small burst of energy is needed to reach equilibrium: in a car engine, the spark plug provides this energy and combustion occurs. Without this burst of energy, the petrol is stable and no combustion occurs (as you will ruefully be aware if you have ever tried to start a car with a flat battery).

The mixture of petrol and oxygen is said to be *thermodynamically* unstable with respect to the products of the reaction, CO_2 and H_2O , but *kinetically* stable. We can be certain that they are thermodynamically unstable because even if the same small energy burst were applied to the products CO_2 and H_2O , they would never convert back to petrol and oxygen.

Kinetically stable means that although the mixture *could* convert to a more stable set of products, it doesn’t do so because an energy barrier separates it from those products. An energy level diagram for a reaction such as the combustion of isooctane is shown below. The products are more stable (lower in energy) than the reactants, but to become the products, the reactants have to overcome a barrier to reaction. This barrier is called the **activation energy** and is usually given the symbol E_a or $\Delta G^\ddagger$.

The differences between E_a and $\Delta G^\ddagger$ need not concern us here; you will find the details in a textbook of physical chemistry.



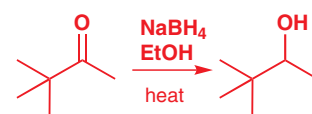
If a reaction cannot proceed until the reactants have sufficient energy to overcome the activation energy barrier, it is clear that, the smaller the barrier, the easier it will be for the reaction to proceed. Likewise, the more energy we give the starting materials in the form of temperature, the more likely it is that they will collide with sufficient combined energy to cross the activation energy barrier. Unlike equilibria, which can change in either direction, reaction rates *always* increase at higher temperatures.

A word of warning, however: heating is not all good for the chemist—not only does it speed up the reaction we want, it will also probably speed up lots of other reactions that we don't want, including perhaps decomposition of the product! We shall see how we can get round this, but first we shall take a closer look at what determines how fast a reaction takes place.

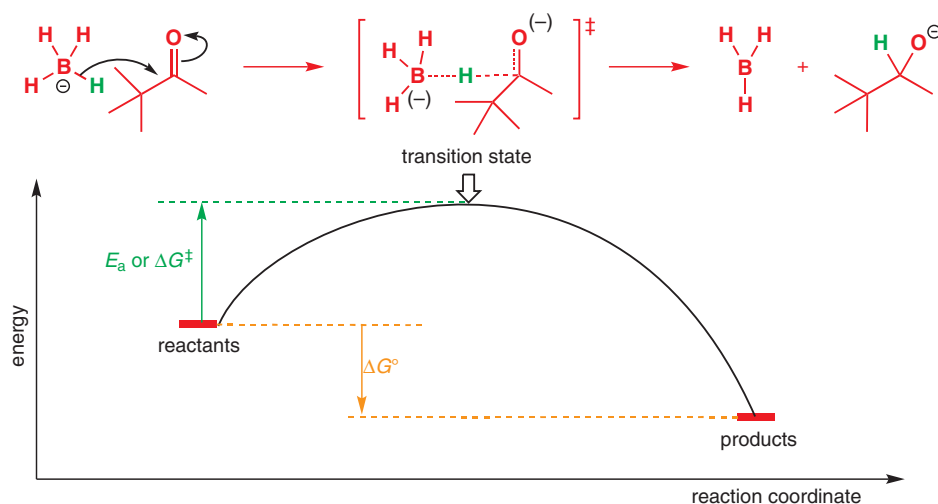
The route from reactants to products: the transition state

The combustion of kinetically stable fuel releases lots of energy by a very complex mechanism. To understand how energy is involved in the progress of a reaction we will need to take a much more simple and familiar mechanism. The reduction of a ketone to an alcohol with sodium borohydride will do. You met this reaction in Chapters 5 and 6 and it should by now be a familiar part of your chemical vocabulary. An example is shown in the margin: in this particular case, the ketone is rather hindered by the adjacent *tert*-butyl group, and the reaction must be heated to form the product. Evidently, then, there is an activation barrier that must be overcome.

Let's think about what that barrier might be. Although the final product is an alcohol, as you know well, the first step is transfer of a hydrogen atom from boron to the carbonyl group, as shown in the mechanism below. Overall, as shown in the energy profile diagram, the products of this step are more stable than the starting materials (ΔG is negative), but to get there the reaction has to pass through the activation energy barrier ($\Delta G^\ddagger$). This barrier—the highest energy point on the profile—must correspond to some structure (which we have shown in square brackets) in which the hydrogen atom is only partly transferred from B to C, and the carbonyl group is only partly broken. We call this structure—the highest energy form through which the molecules must pass to get from reactants to products—the **transition state**. It is often represented in square brackets, frequently with a double dagger symbol $\ddagger$ (to match the activation energy $\Delta G^\ddagger$).



■ To draw a structure representing a transition state is easy: first put in all the bonds that are not affected by the reaction, then use dotted bonds for all those which break or form as the reaction proceeds. You will need to spread charge over appropriate atoms, putting a + or – in brackets to indicate a partial charge.



Notice that the transition state has some features of the reactants and some features of the products. The B–H bond is partly broken, so we represent it as a dotted line, and the new H–C bond is partly formed, so likewise that is dotted too, as is the breaking C=O bond. The negative charge, which starts associated with B and ends on the oxygen atom, is shown in brackets in both locations, to indicate that it is shared between them. It takes energy to get to the transition state because the H has to move away from the B without significant compensation.

Interactive mechanism for borohydride reduction

Conventionally, charges in brackets indicate a significant proportion, usually about 1/2, of a charge, unlike 'δ+' or 'δ–', which might represent only 1/10 or 1/5 of a charge.

But once the transition state is passed, the formation of a stable C–H bond and the migration of a charge to electronegative oxygen means that stability is regained.

A transition state is always unstable and can never be isolated: if the reaction proceeds just a little more forwards or backwards, the energy of the system is lower. Isolating a transition state would be like balancing a marble on top of a bowling ball.

● Transition state

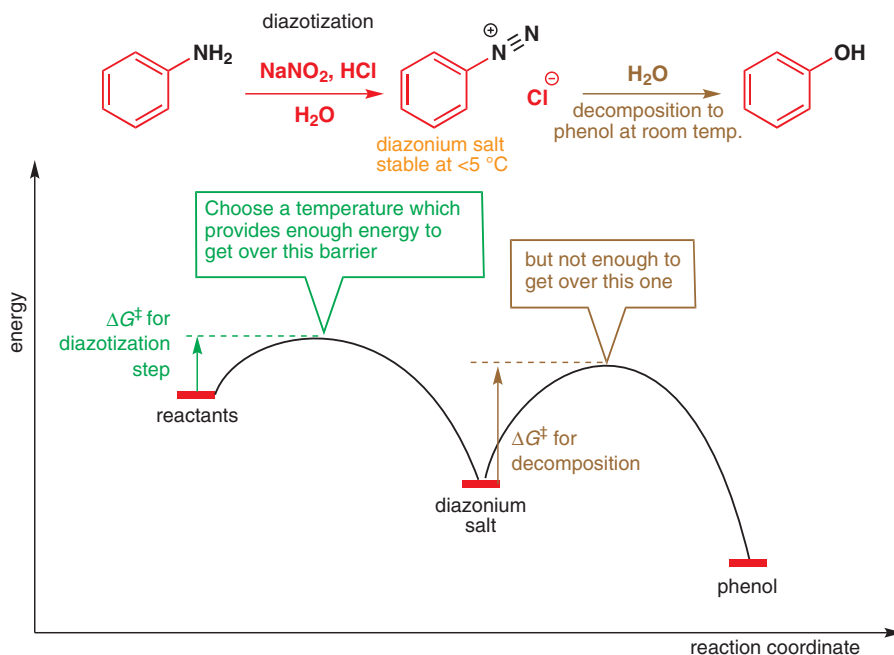
A transition state is a structure that represents an energy maximum on passing from reactants to products. It is not a real molecule in that it may have partially formed or broken bonds and may have more atoms or groups around the central atom than allowed by valence bond rules. It cannot be isolated because it is an energy maximum and any change in its structure leads to a more stable arrangement. A transition state is often shown by putting it in square brackets with a double-dagger superscript.

Why some reactions are done at low temperature

So far in this chapter you have seen that while heating a reaction can change the position of an equilibrium, the usual reason for heating a reaction is to speed it up by giving the reactants more energy to allow them to overcome the activation barrier. But as we said in the introduction, in a typical laboratory you will see many reactions being carried out at low temperatures. Why might a chemist want to *slow a reaction down*?

Well, often molecules can react in several different ways. A good reaction will have a lower activation energy than these alternatives. But often there are other unavoidable reactions waiting in the wings that will compete with the one that is wanted if the molecules have enough energy. The ideal situation is to give the starting materials enough energy to do the reaction we want, but not enough to do anything else: and that means keeping the reaction cold.

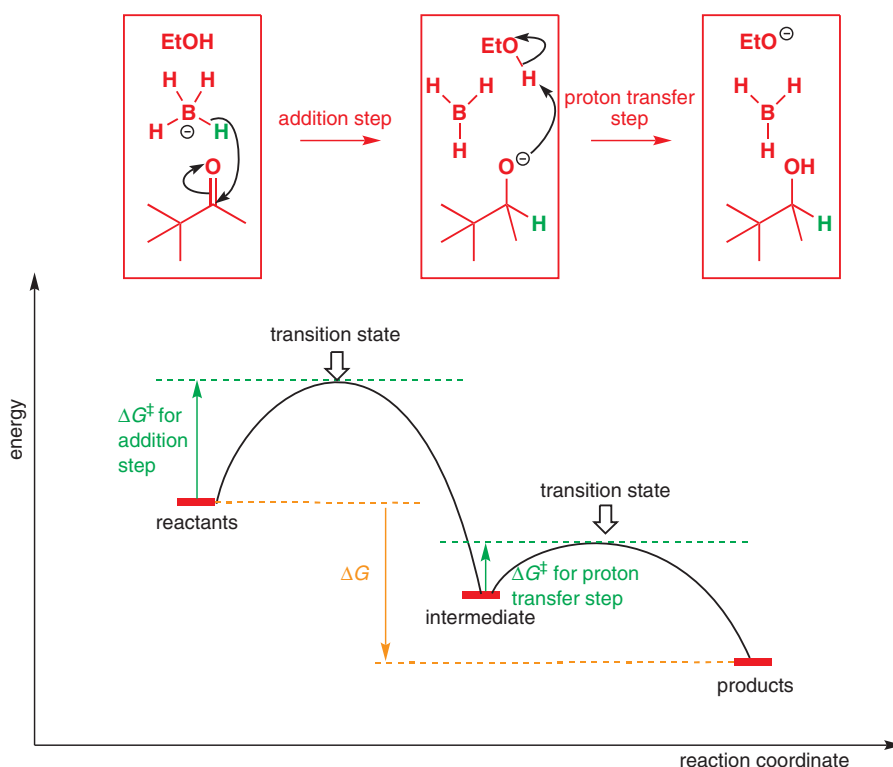
A famous example of a reaction which must be kept cold is the diazotization of anilines to make diazonium salts. The reaction involves treating the amine with nitrous acid (HONO) made from NaNO_2 and HCl. You need not think about the mechanism at this stage—you will meet it in Chapter 22—but the key point is that the product is a rather unstable but very useful diazonium salt. The diazotization takes place readily at room temperature, but unfortunately so does the decomposition of the product to give a phenol. By lowering the temperature, we supply insufficient energy for the phenol formation, but the diazotization still works just fine.



The use of organolithiums (which you saw in Chapter 9) typically involves low temperatures, often as low as $-78\text{ }^{\circ}\text{C}$. Organolithiums are very reactive, and the addition or deprotonation reactions they undergo have activation energies sufficiently low that they proceed even at such temperatures. However, they do also have a tendency to attack some of the solvents which they dissolve best in, such as THF. If lithiations are attempted at higher temperatures, THF also reacts with *s*-BuLi to give the surprising by-products discussed in Chapter 35.

Reaction intermediates

Our mechanism for reduction of the ketone with borohydride is of course not yet complete: there is another step to follow—the protonation of the alkoxide by the ethanol solvent. We can add this step to our energy profile diagram.

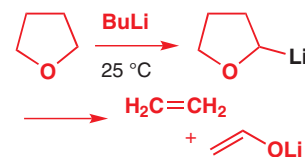


Now the product of the first stage of the reaction is the starting material for the second, which follows on straightaway because the activation energy for the second reaction, the proton transfer, is smaller than for the first one. Notice that we have now labelled the middle set of structures, which includes the alkoxide 'intermediate'. An intermediate is a staging post in a reaction pathway: it is stable for a finite (if short) period. Unlike a transition state it is a minimum rather than a maximum on the reaction energy profile, and therefore has a finite existence—an intermediate *could*, in principle, be isolated, and many have been (particularly at low temperature).

Intermediates and transition states

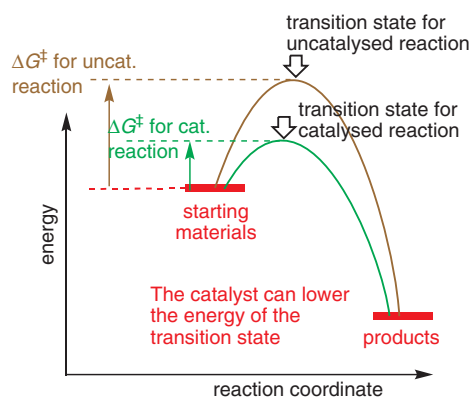
- A **transition state** represents an energy maximum—any small displacement leads to a more stable product. It can never be isolated.
- An **intermediate** is a molecule or ion that represents a localized energy minimum—an energy barrier must be overcome before the intermediate forms something more stable. An intermediate can in principle be isolated (although in practice its high energy can make this difficult).

■ $-78\text{ }^{\circ}\text{C}$ is the convenient temperature of a bath of acetone containing pellets of slowly evaporating solid CO_2 .



Interactive mechanism for borohydride reduction

■ You can think of the diazonium salt on p. 252 as an isolable intermediate en route to the phenol. Notice how the energy profile for that sequence matches the one here.



Catalysis

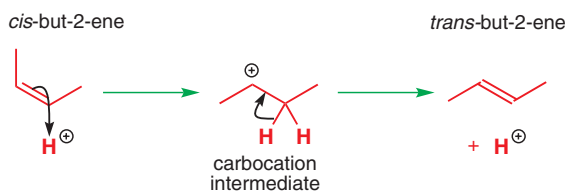
You've met the idea in several places in this book (and probably elsewhere) that a catalyst increases the rate of a reaction. From what you have just read, it must therefore be the case that a catalyst lowers the activation energy for a reaction. It can do this in one or both of two ways: it can lower the energy of the transition state (as shown in the diagram on the left, or it can raise the energy of the starting materials.

- Catalysts work by lowering the activation energy for a reaction.

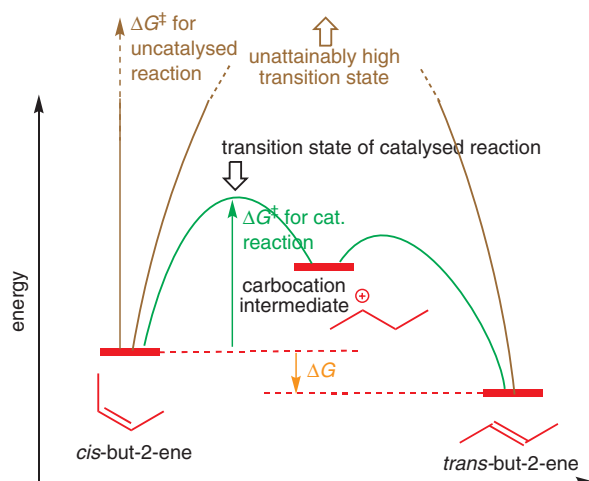
To illustrate the point with one simple example, let's take a reaction which simply does not work without a catalyst: the isomerization of butene. As you saw on p. 105, *cis* but-2-ene is about 2 kJ mol⁻¹ higher in energy than *trans* but-2-ene. This small energy difference would correspond to a 2.2:1, 70:30, *trans*:*cis* ratio of alkenes if they were in equilibrium. But it's a big *if*: the activation energy needed to get from one to the other is of the order of 260 kJ mol⁻¹, which is practically unattainable. A quick calculation predicts that the half-life for the reaction would be approximately 10²⁵ years at room temperature, a time many orders of magnitude longer than the age of the universe. At 500 °C, however, the half-life is a more reasonable 4 hours, but unfortunately, when most alkenes are heated to these sorts of temperatures other unwanted reactions occur.

In order to interconvert the *cis* and *trans* isomers we must use a different strategy: catalysis. You will meet several ways of doing this in Chapter 27, but for the moment we will use just one: catalytic acid. As you saw in Chapter 5, alkenes are nucleophiles, and either isomer of but-2-ene can react with H⁺ from acid to form a transient species known as a carbocation. The activation energy for formation of the carbocation is much less than that for rotation about the C=C bond. The carbocation can now easily lose a proton again, to reform either *cis*- or *trans*-but-2-ene, regenerating the catalyst and allowing the interconversion to take place. Overall, the activation energy is much lower than in the uncatalysed reaction. We will come back to other examples of catalysis later in the chapter.

Acid catalysed isomerization:



You have already seen a case in this chapter, and you met many in Chapter 9, where THF (or diethyl ether) was used as the solvent for organolithium reactions: it coordinates to Li and solubilizes the organometallic compounds. Alcohol solvents cannot be used with organolithiums because they are deprotonated by the strong organolithium bases.



Solvents

The nature of the solvent used in reactions often has a profound effect on how the reaction proceeds. Sometimes, if the solvent is also a reagent, the choice is easy: it's a good idea to carry out hydrolyses of esters in water and formations of esters in the appropriate alcohol because

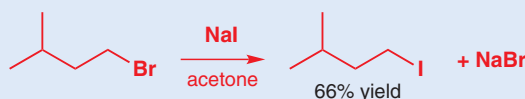
the large concentration of the solvent drives the reaction towards the product, as explained on p. 208. Likewise, the solvent may also catalyse a reaction: ester formation from an acid chloride and an alcohol is often carried out in pyridine as a solvent because pyridine acts as a base catalyst of the reaction (p. 199).

On occasions, the choice of solvent is limited by simple features of the starting materials and products, such as their solubility or reactivity. Simple examples are cases where an inorganic salt is a reagent: ionic compounds are relatively insoluble in most organic solvents. Sodium bromide, for example, dissolves well in water, reasonably well in methanol, a little in ethanol, and hardly at all in most other organic solvents.

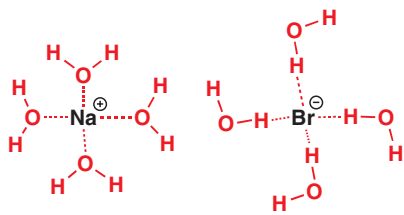
Solubilities of sodium bromide in protic solvents

Solvent	Solubility, g/100 g of solvent
H ₂ O	90
MeOH	16
EtOH	6

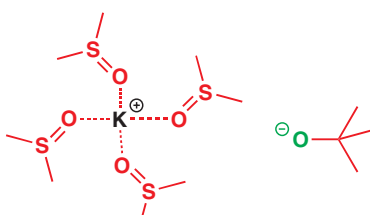
The insolubility of some salts in organic solvents can be used to drive an equilibrium in the direction required. For example, in the synthesis of this alkyl iodide from the alkyl bromide by reaction with sodium iodide, acetone is used as the solvent. Why? Well, sodium iodide is rather more soluble in acetone than is sodium bromide, so as sodium bromide is removed from the equilibrium mixture, more of the starting materials have to convert to the products to restore the equilibrium constant. You will meet more on this reaction in Chapter 15.



Water dissolves sodium bromide well because it solvates both cations and anions: electrostatic interactions with its δ^- oxygen atoms can stabilize the positive sodium ions, while attraction to its δ^+ hydrogen atoms can stabilize the negative bromide ion. Solvents which have polarized bonds like this are known as *polar*. Water and other alcohols are also called protic solvents because they have δ^+ protons that can interact readily with anions.



Water solvates cations and anions



DMSO (a polar aprotic solvent) solvates only cations

Another group of polar solvents lack δ^+ protons: these are the polar aprotic solvents, such as DMSO or DMF. Although they have a localized δ^- at oxygen, which can solvate cations, they are much less good at solvating anions because their molecules do not have a localized accessible δ^+ region. In Chapter 10 (p. 213) you met a specific combination of *t*-BuOK and DMSO to help hydrolyse an amide. This is why DMSO is used here: it solvates the K^+ cation, leaving the *t*-BuO $^-$ unstabilized by solvation. It is desperate to become neutral by finding a proton. Metal alkoxides in DMSO are extremely basic, and when even sodium chloride is dissolved in DMSO the usually innocuous chloride ion becomes quite a powerful nucleophile, as you will see in Chapter 25.

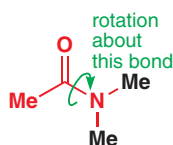
A third group of solvents are not polar at all, but may still dissolve organic molecules quite well. These include hydrocarbons, chlorinated solvents (chloroform), and aromatic solvents (toluene, benzene).

The table below groups common solvents in classes with shared features, and also indicates their polarity. Polarity is measured in various ways—here we give the ‘dielectric constants’—but you do not need to remember the numbers. Learning the general position of a solvent in this sequence of polarity will, however, be a wise investment of your time.

■ Alternatives for chloroform or benzene should be used where possible as there is evidence that these solvents have cancer-causing properties.

Polarity of some common solvents (dielectric constants)

Polar protic solvents		Polar aprotic solvents		Non-polar solvents	
water	80	DMSO	47	chloroform (CHCl ₃)	4.8
methanol	33	DMF	38	diethyl ether	4.3
ethanol	25	acetonitrile	38	toluene	2.4
acetic acid	6	acetone	21	benzene	2.3
		dichloromethane	9.1	cyclohexane	2.0
		tetrahydrofuran (THF)	7.5	hexane	1.9
		ethyl acetate	6.0	pentane	1.8



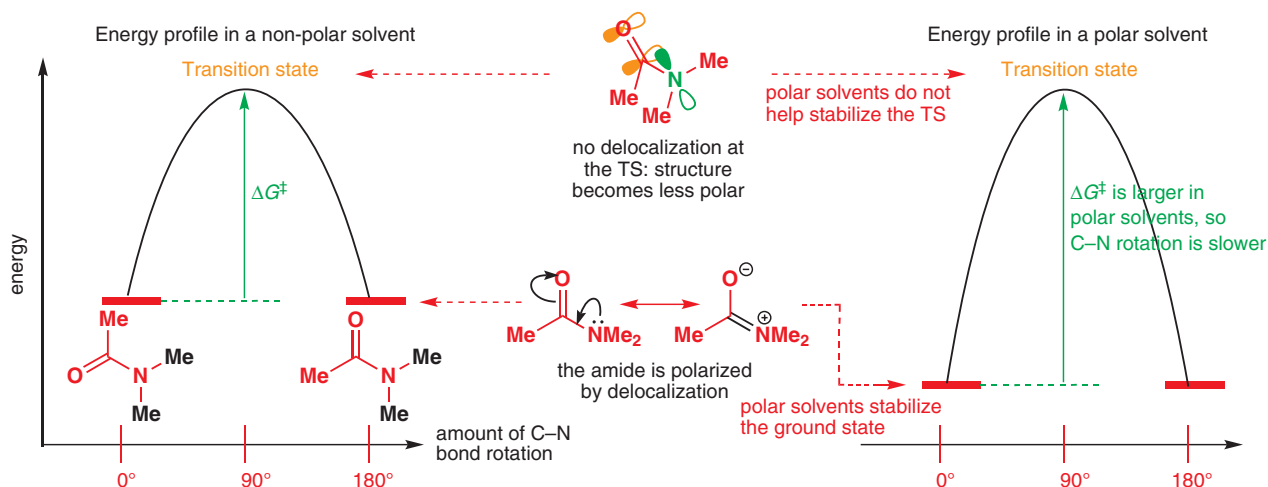
Rate constants for amide C–N bond rotation in dimethylacetamide

Solvent	$\Delta G^\ddagger$, kJ mol ⁻¹
water	80.1
DMSO	76.5
acetone	74.5
cyclohexane	70.0

By virtue of their ability or inability to solvate charged species, solvents can affect the course of reactions by stabilizing or destabilizing a transition state or an intermediate. Here's a very simple example: the effect of solvent on one of the first 'reactions' in this chapter—the rotation about the C–N bond of an amide. The table in the margin shows the activation energy $\Delta G^\ddagger$ for C–N bond rotation in dimethylacetamide (DMA) in a range of solvents. You can immediately deduce that the rate of the rotation is fastest in the least polar solvent, cyclohexane, because the barrier is lowest. Why might this be?

To understand rates, we have to think about activation energies, in other words the difference in energy between the starting materials and the transition state. As you know, an amide in its ground state (in other words, its lowest energy state) is delocalized because of conjugation between the nitrogen's lone pair and the carbonyl group. This delocalization leads to a degree of charge separation and polarization of the amide. But as the C–N bond rotates, the conjugation is broken because the molecule has to pass through a transition state in which the N lone pair is perpendicular to the π system of the carbonyl group. The transition state is therefore less polar than the ground state.

Now, if we compare the effect on this rotation of a non-polar solvent and a polar solvent, this is what will happen. The polar ground state will be stabilized by the polar solvent, and so will be lower in energy, as you see on the right of the diagram below. But the less polar transition state will have about the same energy, whatever the polarity of the solvent. So, in a polar solvent, the amount of energy required to get from the ground state to the transition state (this is the activation energy, E_a or $\Delta G^\ddagger$) is greater than in a non-polar solvent, and bond rotation is slower.



In Chapter 15 you will go on to meet a pair of mechanisms in which the polarity of the transition state is very different. You will now be prepared to expect some very significant solvent effects when such reactions take place.

● **Solvents can affect the rate of a reaction by:**

- participating as a reagent
- acting as a catalyst
- dissolving the reagents
- differentially stabilizing the ground state and transition state.

Rate equations

We've pointed out that reactions go faster at higher temperature because the starting materials have more energy. But temperature is not the sole controller of rate. Two molecules might well collide with plenty of energy, but unless they are two molecules that can actually react, that energy will be lost as heat. Going back again to the reduction of p. 251 (a reminder in the margin), it's obvious that only collisions between ketone (**A**) and borohydride (**B**) get us anywhere—there will be plenty of non-productive collisions between **A** and **A** or **B** and **B**. Obviously the chance of a collision between **A** and **B** is increased the more of each you have, and especially if you have lots of **A** and lots of **B**. In fact, the chance of a successful reaction is proportional to the product of the concentration of **A** and the concentration of **B**. We can express this in a simple *rate equation*:

$$\text{rate of reaction} = k \times [\text{A}] \times [\text{B}]$$

where the value *k* represents the *rate constant* for the reaction. The value of *k* is different for different reactions, and it also varies with temperature. The size of *k* also contains information about how likely it is that the molecules will collide with the right orientation. We call this analysis of the factors affecting the rate of the reactions the **kinetics** of the reaction.

There is of course a link between the activation energy of a reaction and its rate, and the connection between them is known as the Arrhenius equation, after the Swedish chemist Svante Arrhenius (1859–1927) who formulated it and won the Nobel Prize in 1903.

$$k = Ae^{-E_a/RT}$$

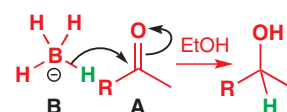
where *k* is the rate constant for the reaction, *R* is the gas constant (see p. 243), *T* is the temperature (in kelvin), and *A* is a quantity known as the pre-exponential factor. Because of the minus sign in the exponential term, the larger the activation energy, *E_a*, the slower the reaction but the higher the temperature, the faster the reaction.

As we discussed on p. 253, the reaction between borohydride and the ketone to make an alkoxide is only the first step of this reaction. Since ethanol likewise has to collide with the alkoxide for this second step to take place, you might very reasonably ask yourself why the rate of formation of the alcohol product does not also depend on [EtOH]: why is the rate equation not

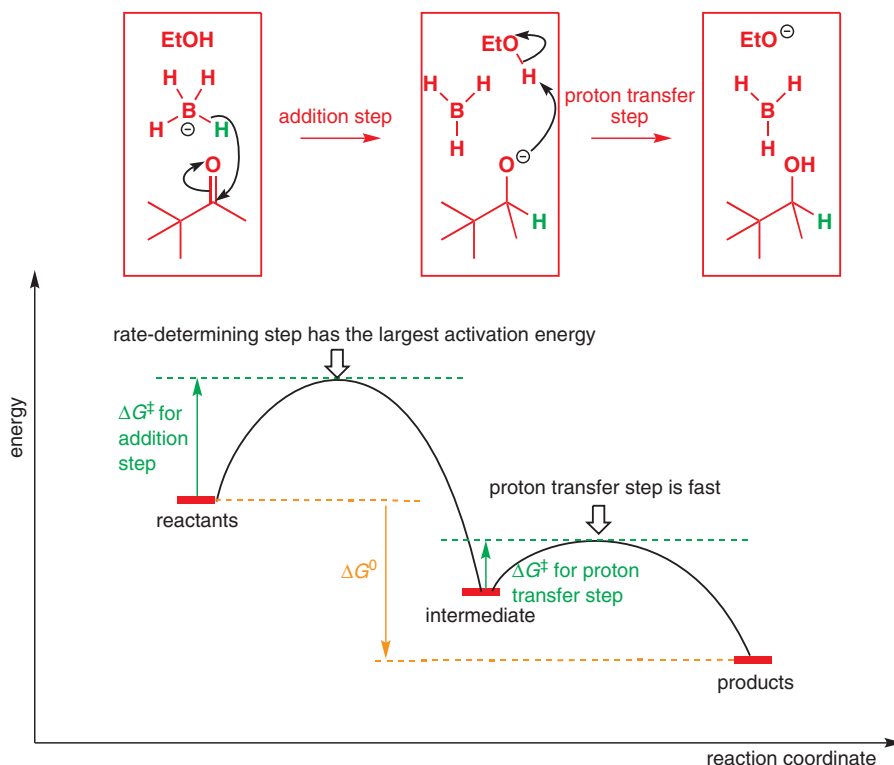
$$\text{rate of reaction} = k \times [\text{ketone}] \times [\text{borohydride}] \times [\text{EtOH}] ?$$

The answer is hinted at in the energy profile diagram you saw on p. 253, which is reproduced below. The activation energy for the proton transfer step is lower than for the addition step, so it happens faster. In fact, it can happen fast *whatever the concentration of ethanol*, so ethanol does not appear in the rate equation. The overall rate of any reaction is determined only by what happens in the mechanistic step that is slowest, known as the **rate-determining step** or **rate-limiting step**. This is a general point about anything that happens in several

➡ We discussed the simple ideas about what must happen for a reaction to take place between two molecules at the start of Chapter 5. We're now adding more detail to those simple concepts.



steps: if you want to empty a football stadium through a set of turnstiles, it is only the rate at which the turnstiles operate that limits the emptying speed—it doesn't matter how quickly or slowly people walk away after they are through.



Proton transfers to or from carbon may be slow.

At several points in Chapters 6, 9, 10, and 11 we have said things like ‘don't worry about the details of the proton transfers’ and now you know why: proton transfers between N and O atoms are fast, and other steps are almost always rate determining. It doesn't really matter how you get a proton from one electronegative atom to another—in reality it will be flitting all over the place and any reasonable route is just as correct as any other.

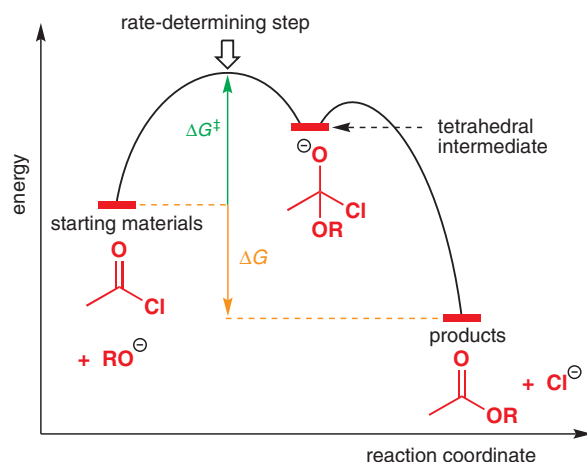
● Proton transfers, particularly between O or N, are always fast and only rarely rate determining.

Kinetics gives us an insight into the mechanism of a reaction

In Chapters 10 and 11 you met some other multistep reactions with intermediates. Take this example: an alkoxide RO^- will react with an acid chloride to form an ester. If we measure how the rate of the formation of the ester varies with the concentration of the alkoxide and of the acid chloride, we discover a rate equation

$$\text{rate} = k[\text{MeCOCl}][\text{RO}^-]$$

Both the acid chloride and alkoxide must therefore be involved in the rate-determining step, which, as you know from Chapter 10, must be the formation of the tetrahedral intermediate. This intermediate is less stable than the starting materials, so the reaction energy profile takes the form shown below, with the highest transition state corresponding to the addition step.



The presence of two species in the rate equation confirms that the reaction is bimolecular (i.e. it involves two molecules), and we call such rate equations *second order*.

Numerous kinetic studies have confirmed that this mechanism, with a tetrahedral intermediate, is the normal pathway by which substitution reactions at carbonyl groups take place, as we explained in Chapter 10. You could draw a similar pathway, and a similar energy profile, for all of the reactions shown on p. 215, adjusting the energies of the starting materials, products, and intermediates appropriately, but all of them are second order, with rate-limiting attack on the carbonyl group.

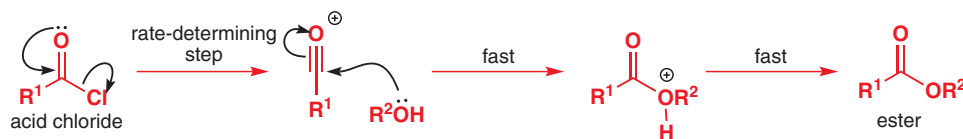
However, there are occasional exceptions. These are not important enough for you to consider them likely when you write substitution mechanisms, but they do illustrate the fact that *kinetics* tells us about *mechanism*.

Here is one: when an acid chloride is heated with an alcohol in the absence of base, an ester forms. However, it turns out that under these conditions the rate equation is first order: it does not matter how much or how little alcohol is added, the rate depends only on the concentration of the acid chloride:

$$\text{rate of reaction} = k[\text{R}^1\text{COCl}]$$

Evidently, from the rate equation, no collision between the acid chloride and the alcohol is required for this reaction to go. The rate-determining step must be *unimolecular*. What actually happens is that the acid chloride decomposes by itself to give a reactive cation with the loss of the good leaving group Cl^- .

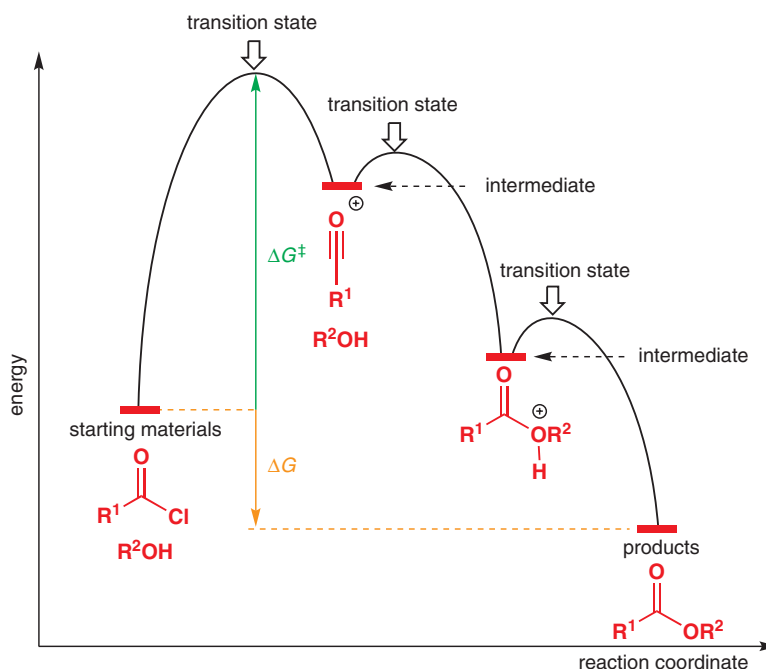
Unusual unimolecular mechanism for ester formation



There are three steps in this reaction scheme, although the last is a trivial deprotonation. The energy barrier must be highest in the first step, which involves the acid chloride alone. The cation is an intermediate (although a short-lived one) with a real existence that reacts rapidly with the alcohol in a step that does not affect the rate of the reaction. The easiest way to picture this detail is in an energy profile diagram:



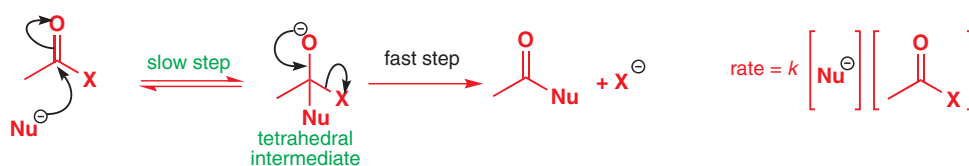
■ This is not the recommended way to make an ester: the reaction is much better if a base is added, in which case it follows the usual addition-elimination mechanism.



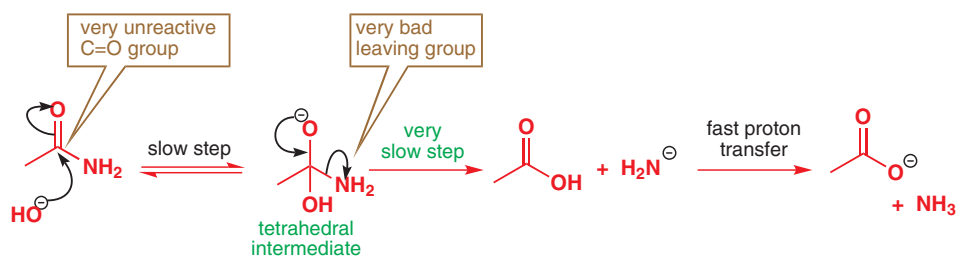
Notice that the products are again lower in energy than the starting materials, and although there are three transition states in this reaction, only the highest-energy transition state (the first one here) matters in determining the reaction rate. The reaction now passes through two intermediates (local minima). It is often the case that when intermediates are involved in a reaction, the highest-energy transition state is associated with the formation of the highest-energy intermediate.

What does third-order kinetics mean?

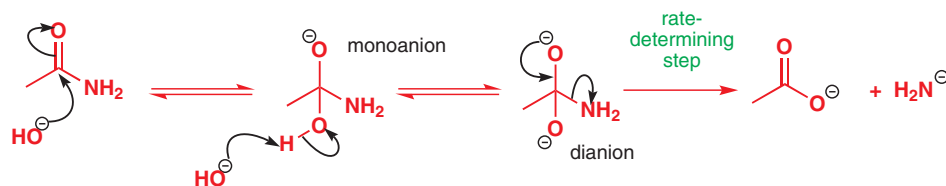
The first-order kinetics of this unusual substitution reaction is here to illustrate a point, but it should not distract you from the fact that most nucleophilic substitutions of carboxylic acid derivatives (the reactions you met in Chapter 10) are bimolecular reactions with rate-determining formation of the tetrahedral intermediate.



However, something different again happens when we come to reactions of amides. Because of the delocalization of the nitrogen lone pair into the carbonyl group, nucleophilic attack on the carbonyl group is very difficult. In addition, the leaving group (NH_2^- , with $\text{p}K_a$ of NH_3 about 35) is very bad indeed.



What happens as a consequence is that in the hydrolysis of amides the second step—the breakdown of the tetrahedral intermediate—becomes rate determining. But this offers the opportunity for base catalysis of this step. If a second hydroxide ion removes the proton from the tetrahedral intermediate, the loss of NH_2 from what is now a dianion is made easier, and a stable carboxylate ion is formed directly.



➡ In Chapter 10 (p. 213) you met a method for hydrolysing amides that exploits this second deprotonation.

Notice that in the first mechanism just one hydroxide ion is involved, whereas now two are involved: one is consumed to form product, but the second is in fact regenerated when the product NH_2^- anion reacts with water—in other words the second hydroxide ion is a catalyst.

The rate equation for the amide hydrolysis reflects this involvement of two hydroxide ions: the rate depends on the square of the hydroxide ion concentration and it is *third order*. We'll label the rate constant k_3 to emphasize this:

$$\text{rate} = k_3[\text{MeCONH}_2] \times [\text{HO}^-]^2$$

But you may be asking yourself where this third-order kinetics comes from, since the hydroxide ions are not actually involved in the rate-determining step. In fact, third-order kinetics hardly ever mean the real simultaneous termolecular collision of three molecules at once—such events are just too rare.

The rate-determining step here is actually unimolecular—the collapse of the dianion. So we expect

$$\text{rate} = k[\text{dianion}]$$

We don't know the concentration of the dianion but we do know that it's in equilibrium with the monoanion—we'll call this equilibrium constant K_2 :

$$K_2 = \frac{[\text{dianion}]}{[\text{monoanion}][\text{HO}^-]}$$

and so $[\text{dianion}] = K_2[\text{monoanion}][\text{HO}^-]$.

This sort of helps, but we still don't know what $[\text{monoanion}]$ is, other than that it's again in equilibrium, this time with the amide—we'll call this equilibrium constant K_1 :

$$K_1 = \frac{[\text{monoanion}]}{[\text{amide}][\text{HO}^-]}$$

and so $[\text{monoanion}] = K_1[\text{amide}][\text{HO}^-]$.

Substituting these values in the simple rate equation we discover that

$$\text{rate} = k[\text{dianion}] \text{ becomes}$$

$$\text{rate} = kK_1K_2[\text{amide}][\text{HO}^-]^2$$

The third-order kinetics result from two equilibria starting with the amide and involving two hydroxide ions, followed by a unimolecular rate-determining step, and the 'third-order rate constant' k_3 is actually a product of the two equilibrium constants and a first-order rate constant:

$$k_3 = k \times K_1 \times K_2$$

This often happens with reactions with late rate-determining steps: the rate constant can depend on the concentrations of any species involved before the rate-determining step (although not necessarily in that step itself) but never depends on species involved after the rate-determining step.

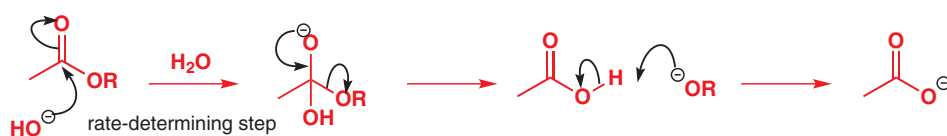
■ In Chapter 39 we will discuss in much more detail how such experiments are designed; much sooner, in Chapter 15, you will meet another pair of mechanisms—one first order and one second order—that tell us a lot about the reactivity of the molecules involved.

Just because a proposed mechanism gives a rate equation that fits the experimental data, it does not necessarily mean that it is the *right* mechanism; all it means is that it is consistent with the experimental facts so far, but there may be other mechanisms that also fit. It is then up to the experimenter to design cunning experiments to try to rule out other possibilities.

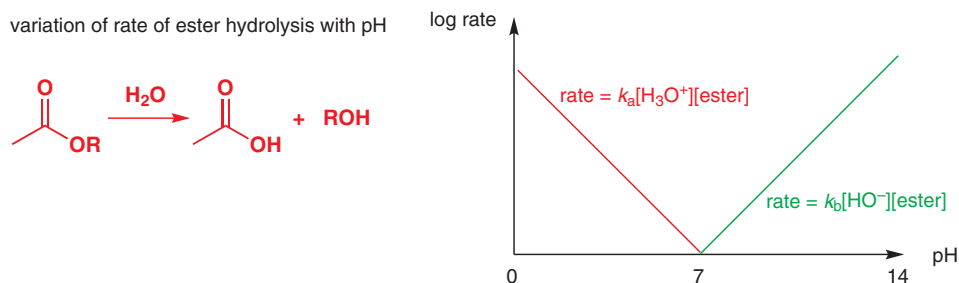
Mechanisms are given throughout this book—eventually you will learn to predict what mechanism to expect for a given type of reaction, but this is because earlier experimentalists have worked out the mechanisms by a study of kinetics and other methods.

Catalysis in carbonyl substitution reactions

The amide hydrolysis you have just met is much faster in base because base (in this case hydroxide) deprotonates the intermediate and makes it more reactive. The same is true for many other base-catalysed processes: often it is the nucleophile that is made more reactive by deprotonation to form an anion. For example, ester hydrolysis is faster at higher pH because the higher the pH the more hydroxide there is to act as a nucleophile.

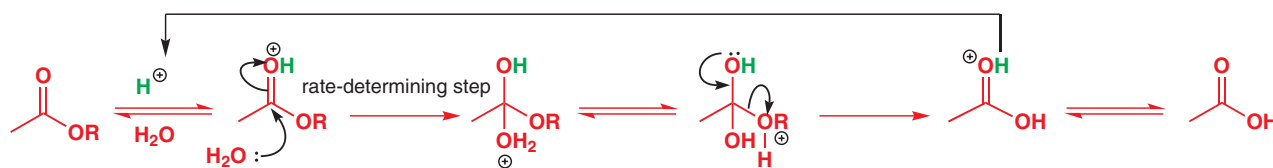


We can plot this on a graph of rate vs. pH:



The rate equation at high pH is second order, as you expect, and depends on the concentration of hydroxide and the concentration of the ester. Notice, though, that below pH 7 the rate starts to increase again as the concentration of $[H^+]$ increases. This is because ester hydrolysis is also acid catalysed, as you saw in Chapter 10. At acidic pH, a new mechanism takes over in which protonation of the carbonyl group accelerates attack of weakly nucleophilic water.

■ Notice that the acid is truly catalytic as the proton (coloured green) is recycled.



■ You will also see rate constants labelled in other ways—this is a matter of choice. A common method is to use k_1 for first-order, k_2 for second-order, and k_3 for third-order rate constants, for example.

The reaction is still bimolecular but the rate constant is different: we can represent the two processes by two rate equations, labelling the rate constants k_a and k_b with the suffixes 'a' for acid and 'b' for base to show more clearly what we mean:

$$\text{rate of ester hydrolysis in acid (pH} < 7) \text{ solution} = k_a[\text{MeCO}_2\text{R}][\text{H}_3\text{O}^+]$$

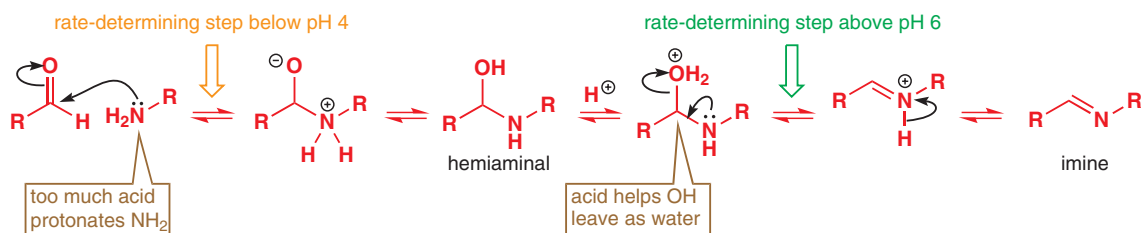
$$\text{rate of ester hydrolysis in basic (pH} > 7) \text{ solution} = k_b[\text{MeCO}_2\text{R}][\text{HO}^-]$$

This is typical acid–base catalysis, known as ‘specific acid–base catalysis’ because the specific acid and base involved are H^+ (or H_3O^+) and OH^- . The form of the pH dependence of the rate tells us that there is a choice of two mechanisms—the one that is faster is the one that is observed.

You met a reaction in Chapter 11 whose rate has a very different pH dependence: imine formation. To remind you, here is the mechanism again. We pointed out in Chapter 11 that

➡ This mechanism is discussed on p. 231.

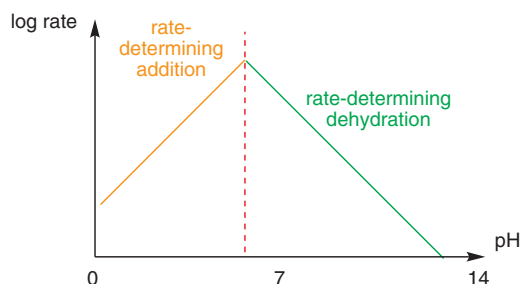
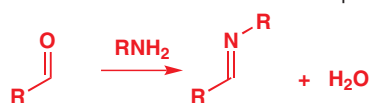
the reaction is acid catalysed because acid is needed to help water leave. But too much acid is a problem because it protonates the starting amine and slows the reaction down.



For these reasons, the pH-rate profile for imine formation looks like this: there is a maximum rate around pH 6, and either side the reaction goes more slowly.

 Interactive mechanism for imine formation

variation of rate of imine formation with pH



The difference now is that at low pH, the rate-determining step changes from being the dehydration step (which can then go very fast because of the high concentration of acid) to being the addition step, which is slowed down by protonation of the amine. Whereas a reaction will always go by the fastest of the available mechanisms, it is also bound to go at the rate of the slowest step in that mechanism.

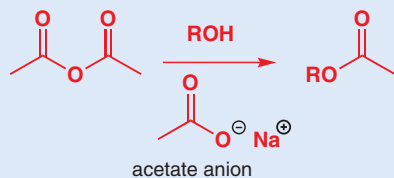
● Multistep reaction rates

The overall rate of a multistep reaction is decided by:

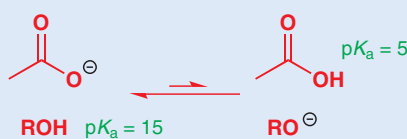
- the *fastest* of the available mechanisms
- the *slowest* of the possible rate-determining steps.

Catalysis by weak bases

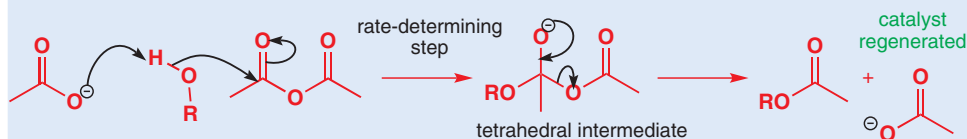
In Chapter 10 we used pyridine as a catalyst in carbonyl substitution reactions, even though it is only a weak base. Catalysis by pyridine involves two mechanisms, and is discussed on p. 200. Acetate ion is another weak base which can catalyse the formation of esters from anhydrides:



acetate is too weak to deprotonate ROH:



The problem is, it is far too weak a base (acetic acid has a pK_a of 5) to deprotonate the alcohol (pK_a 15), so it can't be forming alkoxide (in the way that hydroxide would for example). But what it can do is to remove the proton from the alcohol as the reaction occurs.



This type of catalysis, which is available to any base, not only strong bases, is called *general base catalysis* and will be discussed more in Chapter 39. It does not speed the reaction up very much but it does lower the energy of the transition state leading to the tetrahedral intermediate by avoiding the build-up of positive charge as the alcohol adds. The disadvantage of general base catalysis is that the first, rate-determining, step really is termolecular (unlike in the amide hydrolysis mechanism you met on p. 261). It is inherently unlikely that three molecules will collide with each other simultaneously. In this case, however, if ROH is the solvent, it will always be nearby in any collision so a termolecular step is just about acceptable.

Kinetic versus thermodynamic products

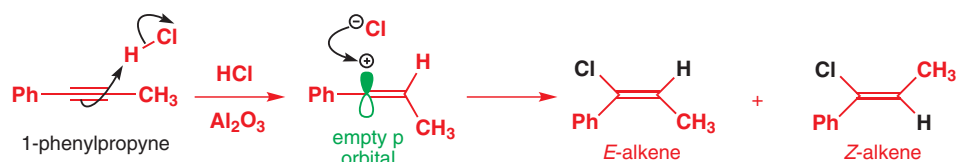
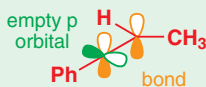
We started this chapter with a discussion of thermodynamics: the factors that govern equilibria. We then moved onto rates: the factors that determine the *rate* at which reactions proceed. Depending on the reaction, either may be more important, and in general:

- Reactions under **thermodynamic control** have outcomes that depend on the **position of an equilibrium** and therefore the relative stability of the possible products.
- Reactions under **kinetic control** have outcomes that depend on the rate at which the reaction proceeds, and therefore on the **relative energies of the transition states** leading to the alternative products.

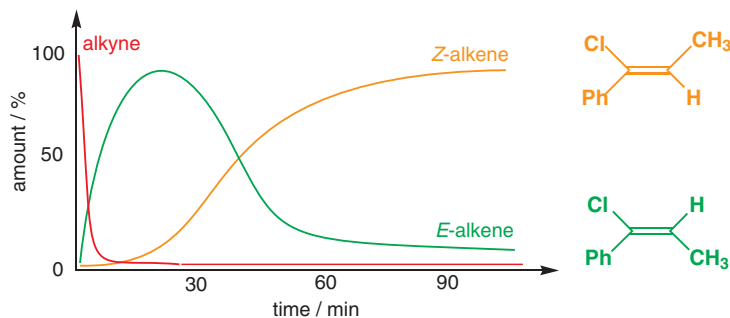
➡ There are further examples of contrasting kinetic and thermodynamic control in Chapters 19 and 22.

■ Hydrogen chloride is a gas, but it can be absorbed onto the surface of the alumina for convenient handling.

■ It's worth taking a moment to think about the structure of the intermediate cation here: the cationic carbon is sp hybridized (linear) with an empty p orbital perpendicular to the p orbitals of the double bond (it is the p orbital that used to be involved in the second π bond of the alkyne).



The two alkenes are labelled *E* and *Z*. After about 2 hours the main product is the *Z*-alkene. However, this is not the case in the early stages of the reaction. The graph below shows how the proportions of the starting material and the two products change with time.



Points to note:

- When the alkyne concentration drops almost to zero (10 minutes), the only alkene that has been formed is the *E*-alkene.

- As time increases, the amount of *E*-alkene decreases as the amount of *Z*-alkene increases.
- Eventually, the proportions of *E*- and *Z*-alkene do not change.

Since it is the *Z*-alkene that dominates at equilibrium, this must be lower in energy than the *E*-alkene. Since we know the ratio of the products at equilibrium, we can work out the difference in energy between the two isomers:

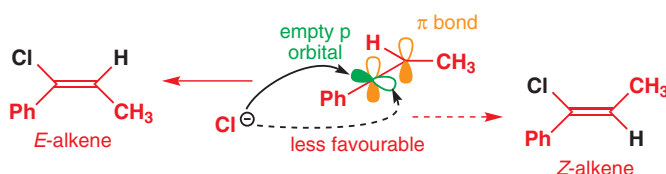
ratio of *E*:*Z*-alkenes at equilibrium = 1:35

$$K_{\text{eq}} = \frac{[Z]}{[E]} = 35$$

$$\Delta G = -RT \ln K = -8.314 \times 298 \times \ln(35) = -8.8 \text{ kJ mol}^{-1}$$

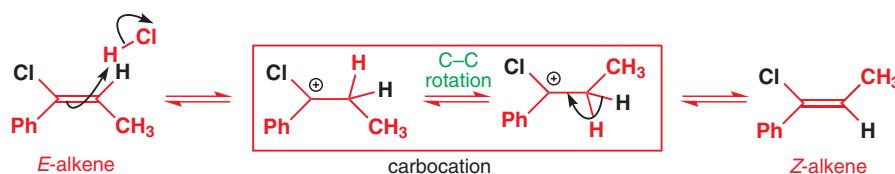
that is, the *Z*-alkene is 8.8 kJ mol⁻¹ lower in energy than the *E*-alkene.

However, although the *Z*-alkene is more stable, the *E*-alkene is formed faster under these conditions: the route to the *E*-alkene must have a smaller activation energy barrier than *trans* addition. This is quite easy to understand: the intermediate cation has no double-bond geometry because the cationic C is sp hybridized (linear). When chloride attacks, it prefers to attack from the side of the H atom rather than the (bigger) methyl group.

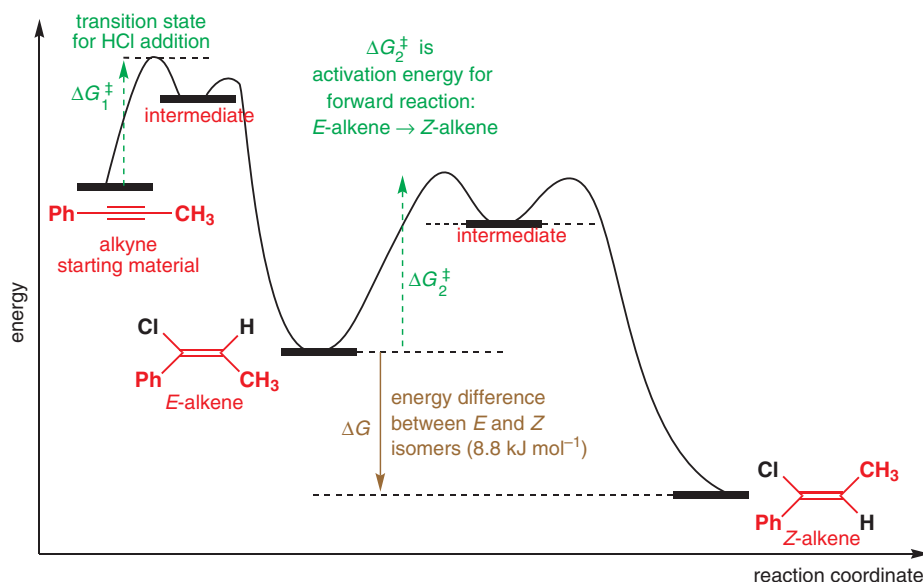


■ You might normally expect an *E*-alkene to be more stable than a *Z*-alkene—it just so happens here that Cl has a higher priority than Ph and the *Z*-alkene has the two largest groups (Ph and Me) *trans* (see p. 392 for rules of nomenclature).

There must then be some mechanism by which the quickly formed *E*-alkene is converted into the more stable *Z*-alkene. The conditions are acidic, so the most likely mechanism is the acid-catalysed alkene isomerization you saw earlier in the chapter:



This information can be summarized on an energy profile diagram:



Initially, the alkyne is converted into the *E*-alkene via the intermediate linear cation. The activation energy for this step is labelled $\Delta G_1^\ddagger$. The *E*-alkene can convert to the *Z* isomer via an intermediate, with activation energy $\Delta G_2^\ddagger$. Since $\Delta G_1^\ddagger$ is smaller than $\Delta G_2^\ddagger$, the *E*-alkene forms faster than it isomerizes, and all the alkyne is rapidly converted to the *E*-alkene. But over the course of the reaction, the *E*-alkene slowly isomerizes to the *Z*-alkene. An equilibrium is eventually reached that favours the *Z*-alkene because it is more stable (by 8.8 kJ mol^{-1} , as we calculated earlier). Why doesn't the *Z*-alkene form faster than the *E*? Well, as we suggested above, the transition state for its formation from the linear cation must be higher in energy than the transition state for formation of the *E*-alkene, because of steric hindrance.

● Kinetic and thermodynamic products

- The *E*-alkene is formed faster and is known as the **kinetic product** or the **product of kinetic control**.
- The *Z*-alkene is more stable and is known as the **thermodynamic product** or the **product of thermodynamic control**.

If we wanted to isolate the kinetic product, the *E*-alkene, we would carry out the reaction at low temperature and not leave it long enough for equilibration. If, on the other hand, we want the thermodynamic product, the *Z*-alkene, we would leave the reaction for longer at higher temperatures to make sure that the larger energy barrier yielding the most stable product can be overcome.

Summary of mechanisms from Chapters 6–12

In Chapter 5 we introduced basic arrow-drawing. A lot has happened since then and this is a good opportunity to pull some strands together. You may like to be reminded:

1. When molecules react together, one is the *electrophile* and one is the *nucleophile*.
2. Electrons flow from an electron-rich to an electron-poor centre.
3. Charge is conserved in each step of a reaction.

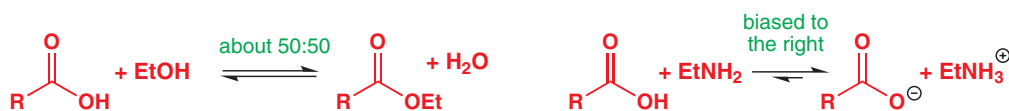
These three considerations will help you draw the mechanism of a reaction that you have not previously met.

Types of reaction arrows

1. Simple reaction arrows showing that a reaction goes from left to right or right to left.



2. Equilibrium arrows showing the extent and direction of equilibrium.

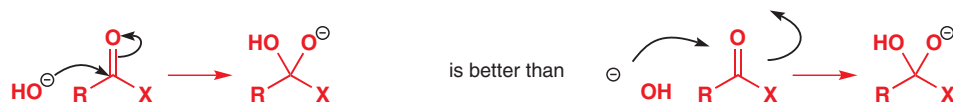


3. Delocalization or conjugation arrows showing two different ways to draw the same molecule. The two structures ('canonical forms' or 'resonance structures') must differ only in the position of electrons.

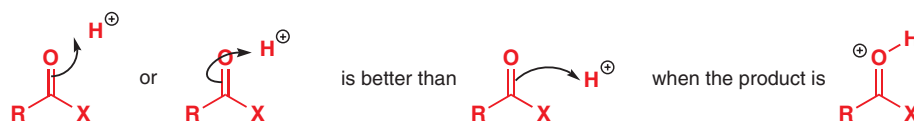


Using curly arrows

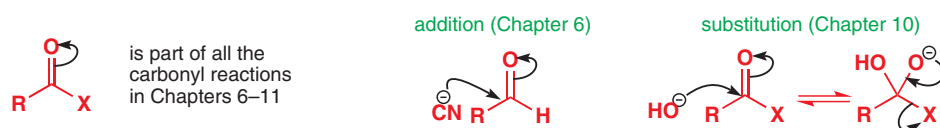
1. The curly arrow should show clearly where the electrons come from and where they go to.



2. If electrophilic attack on a π or σ bond leads to the bond being broken, the arrows should show clearly which atom bonds to the electrophile.

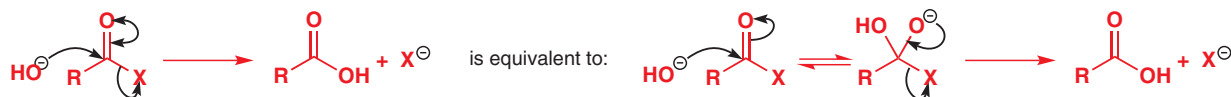


3. Reactions of the carbonyl group are dominated by the breaking of the π bond. If you use this arrow first on an unfamiliar reaction of a carbonyl compound, you will probably find a reasonable mechanism.

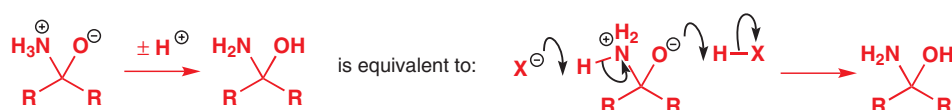


Shortcuts in drawing mechanisms

1. The most important is the double-headed arrow on the carbonyl group used during a substitution reaction.



2. The symbol $\pm H^+$ is shorthand for the gain and loss of a proton in the same step (usually involving N, O, or S: such steps are usually kinetically very fast).



Further reading

For a more in-depth description of reaction pathways, see J. Keeler and P. Wothers, *Why Chemical Reactions Happen*, OUP, Oxford, 2003.

A physical chemistry text such as *Physical Chemistry*, 9th edn, by P. Atkins and J. de Paula, OUP, Oxford, 2011, will give you much more mathematical detail.

An excellent modern and rather more advanced physical organic book is E. V. Anslyn and D. A. Dougherty, *Modern Physical Organic*

Chemistry, University Science Books, South Orange New Jersey, 2005.

Equilibrium constants for hemiacetal formation: J. P. Guthrie *Can J. Chem.* 1975, 898.

Solvent dependence of bond rotation in amides: T. Drakenberg, K. I. Dahlqvist, and S. Forsen *J. Phys. Chem.*, 1972, 76, 2178.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>